CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 4



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Recap of Previous Lecture



Topic Pipeline Hazards

Topic Data Hazard

Topic Operand Forwarding

Topics to be Covered



- **Topic** Data Hazard Classification
- **Topic** RAW, WAW, WAR
- **Topic** CPI In Pipeline



Topic : Hazard Classification

Assume that there are two instructions i and j, and i is executed before j

1. RAW
2. WAW
3. WAR



Topic : Read After Write (RAW)

→ true dependency

j tries to read a source before *i* writes it. So *j* incorrectly gets the old value.

i : R1 ← R2 + R3

j : R5 ← R1 * R6

Soln :- operand forwarding



Topic : Write After Write (WAW)

→ write data dependency

→ false dependency

j tries to write an operand before it is written by i

i : $R1 \leftarrow R2 + R3$

j : $R1 \leftarrow R5 * R6$

} Soln :- Register Renaming (H/w Soln)

⇓

done a run time

$R1 \leftarrow R2 + R3$

$R8 \leftarrow R5 * R6$



Topic : Write After Read (WAR)

→ anti dependency

→ False dependency

j tries to write an operand before i reads it, hence i incorrectly gets new value

i : $R1 \leftarrow R2 + R3$

j : $R2 \leftarrow R5 * R6$

Soln :-

Reg. Renaming

i : $R1 \leftarrow R2 + R3$

j : $R9 \leftarrow R5 * R6$

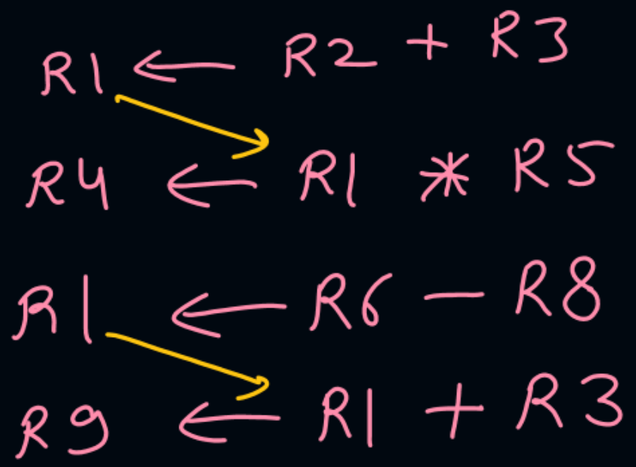
[NAT]

#Q. Count the number of RAW, WAW and WAR dependencies? ⇒ 4, 1, 3

ADD R2, R1, R0	1. R2←R1+R0
MUL R4, R3, R2	2. R4←R3*R2
SUB R6, R5, R4	3. R6←R5-R4
ADD R6, R7, R8	4. R6←R7+R8
MUL R7, R1, R2	5. R7←R1*R2
SUB R1, R3, R4	6. R1←R3-R4

There is RAW dependency b/w inst^n 1 and 2 for R2

ex 3- no. of RAW dependencies ?



Ans = 2

[MSQ] GATE - PYQ

#Q. An instruction format has the following structure

Instruction number: opcode destination reg, source reg-1, source reg-2

Consider the following sequence of instruction to be executed in a pipelined processor

I1: DIV R3, R1, R2 — R3 ← R1 / R2

I2: SUB R5, R3, R4 — R5 ← R3 − R4

I3: ADD R3, R5, R6 — R3 ← R5 + R6

I4: MUL R7, R3, R8 — R7 ← R3 * R8

Which of the following statement is/are TRUE?

A. There is a WAW dependency on R3 between I3 and I4

B. ✓ There is a WAW dependency on R3 between I1 and I3

C. ✓ There is a RAW dependency on R3 between I1 and I2

D. ✓ There is a RAW dependency on R3 between I2 and I3

[MCQ]

#Q. For a pipelined CPU with a single ALU, consider the following situations

I. The J + 1st instruction uses the result of the Jth instruction as an operand → data dependency

II. The execution of a conditional jump instruction → control hazard

III. The Jth and J + 1st instructions require the ALU at the same time → structural hazard

Which of the above can cause a hazard

A. I and II only
B. II and III only
C. III only
D. ✓ All the three

#Q. Register renaming is done in pipelined processors:

a) as an alternative to register allocation at compile time
b) for efficient access to function parameters and local variables
c) to handle certain kinds of hazards
d) as part of address translation

[NAT]

Ans. 10, 14

#Q. Consider the following sequence of instructions, which is to be executed on a 5 stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding $= 5 + 6 - 1 = 10$
2. When pipeline does not use operand forwarding

$5 + 6 - 1 = 10$

stalls due to data hazard $= 2 * (5 - 3)$

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ADD R2, R1, R0	R2←R1+R0
MUL R4, R3, R2	R4←R3*R2
SUB R6, R5, R4	R6←R5-R4
ADD R6, R7, R8	R6←R7+R8
MUL R7, R1, R2	R7←R1*R2
SUB R1, R3, R4	R1←R3-R4

[NAT]

H.W.

#Q. Consider the following sequence of instructions, which is to be executed on a 5 stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding
2. When pipeline does not use operand forwarding

ADD R2, R1, R0	R2←R1+R0
MUL R4, R3, R2	R4←R3*R2
LOAD R7, (1000)	R7←M[1000]
ADD R6, R7, R8	R6←R7+R8
MUL R7, R1, R2	R7←R1*R2
SUB R1, R3, R4	R1←R3-R4



Topic : CPI Calculation in Pipeline



without hazard :-

for n inst^{ns} no. of cycles $= k + n - 1$

for 1 inst^{n} —— // —— $(CPI_{avg}) = \dfrac{k+n-1}{n}$

Ideal condition $\Rightarrow$

$$CPI_{avg} = 1$$

with hazard :-

assume due to all type of hazard, total stall cycles = x

for n inst^{ns}, no. of cycles $= (k+n-1) + x$

for 1 inst^{n}, —— // —— $(CPI_{avg}) = \dfrac{k+n-1+x}{n}$

In ideal condition :-

$$CPI_{avg} = \frac{n+x}{n} = \frac{1}{} + \frac{x}{n}$$

Avg. instⁿ execution time = CPI * t_p

[NAT]

#Q. Consider a 5-stage pipeline which is executing a program of 1000 instructions.

Among all instructions 200 instructions cause 2 stall cycles each.

1. Calculate CPI of pipeline?
2. If pipeline cycle time is 3ns then what is average instruction execution time?
3. Calculate CPI of pipeline in ideal conditions?
4. If pipeline cycle time is 3ns then what is average instruction execution time in ideal conditions?

total stalls (x) = $200 * 2 = 400$

1. $CPI = \frac{5 + 1000 - 1 + 400}{1000} = 1.404$

2. $A.I.E.T. = 1.404 * 3\,ns = 4.212\ ns$

3. $CPI_{ideal} = 1 + \frac{400}{1000} = 1.4$

4. A.I.E.T. in ideal condition $= 1.4 * 3\,ns = 4.2\ ns$

CPI Calculations :-

out of 1000, 200 instns cause stalls

$\Downarrow$

$\frac{200}{1000} = 0.2$ or 20% instns cause stalls

$$CPI = 1 + \frac{400}{1000}$$

$$= 1 + \frac{200 * 2}{1000}$$

$$= 1 + 0.2 * 2$$

Ques) If 30% instns incur 3 stalls each.

CPIavg = 1.9 ?

Soln

CPIavg = 1 + 0.3 * 3 = 1.9

Ques) If 20% instns incur 2 stalls each, and 15% instns incur 3 stalls each

CPIavg = 1.85 ?

Soln

CPIavg = 1 + 0.2 * 2 + 0.15 * 3
= 1.85

Speed up :-

without hazard, in ideal condition $\quad \text{Speed up} = \dfrac{t_n}{1 * t_p}$

with hazard, ——— 11 ——— $= \dfrac{t_n}{CPI_{avg} * t_p}$

[NAT]

of 4ns.

#Q. Consider a 5-stage instruction pipeline, where all stages take equal delay. When an application is executing on this 5-stage pipeline, consider 20% of the instructions incur 3 pipeline stall cycles is.

1. Average CPI for pipeline? $1 + 0.2 * 3 = 1.6$
2. Average instruction execution time for pipeline? $1.6 * 4ns = 6.4ns$
3. The speedup of pipeline achieved with respect to non-pipeline?

$t_n = 4 + 4 + 4 + 4 + 4 = 20ns$

$$\text{Speed up} = \frac{20ns}{6.4ns} = 3.125$$

Ques) In prev. questn speed up of pipeline without hazard?

$$S = \frac{20}{4} = 5$$

Ques) In prev. questn efficiency of pipeline?

Soln

$$= \frac{3.125}{5}$$

$$= 0.625 \text{ or } 62.5\%$$

$$\text{Efficiency of pipeline} = \frac{\text{achieved speed up}}{\text{max speed up}}$$

[NAT]

#Q. Consider a 5-stage instruction pipeline, where stages take delays 5ns, 4ns, 6ns, 4ns and 5ns respectively. The pipeline is used to execute a program in which 25% instructions cause 4 stalls due to hazard. The average instruction execution time in the pipeline is __12__ ns?

$t_p = \max(5,4,6,4,5) = 6\,ns$

$CPI = 1 + 0.25 * 4$

$= 2$

$A.I.E.T. = 2 * 6\,ns$

$= 12\,ns$

[NAT] GATE-PYQ

$t_p = \frac{1}{1GHz} = 1\,ns$

#Q. A CPU has a five-stage pipeline and runs at 1 GHz frequency. Instruction fetch happens in the first stage of the pipeline. A conditional branch instruction computes the target address and evaluates the condition in the third stage of the pipeline. The processor stops fetching new instructions following a conditional branch until the branch outcome is known. A program executes 10^9 instructions out of which 20% are conditional branches. If each instruction takes one cycle to complete on average, the total execution time of the program is __1.4__ seconds?

stalls due to each branch inst^n = 3 − 1 = 2

$CPI = 1 + 0.2 * 2 = 1.4$

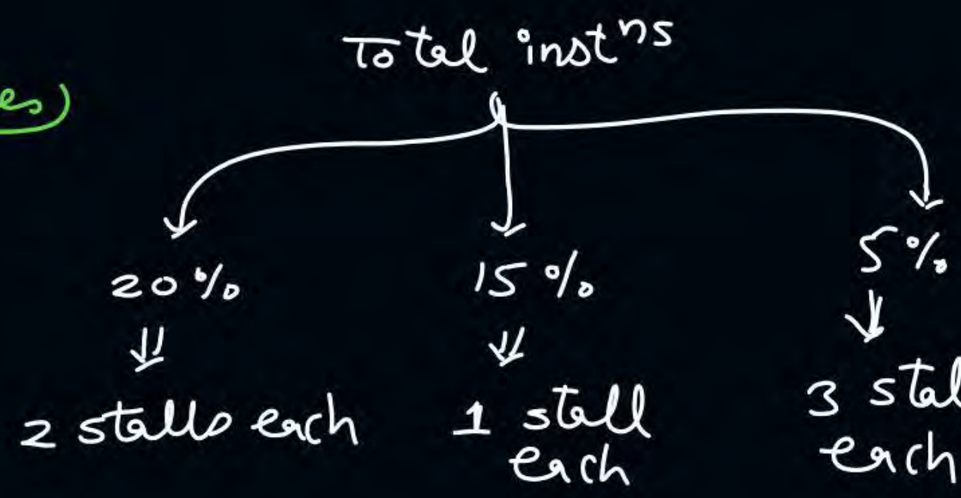
$A.I.E.T. = 1.4 * t_p$

$= 1.4 * 1\,ns = 1.4\,ns$

Prog. execution time $= 1.4\ ns * 10^9$

$= 1.4\ sec$

Ques)

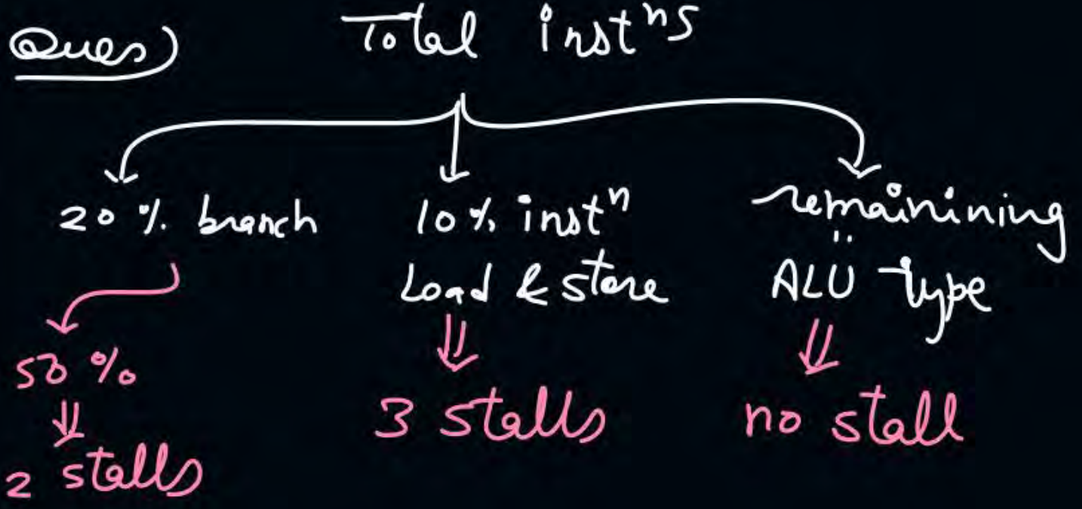


CPI = 1.7 ?

Soln :- $1 + (0.2 * 2) + (0.15 * 1) + (0.05 * 3)$

$= 1.7$ Ans.

Ques)



CPI = 1.5 ?

Soln $= 1 + (0.2 * 0.5 * 2) + (0.1 * 3)$

$= 1.5$ Ans.



2 mins Summary

Topic Data Hazard Classification

Topic RAW, WAW, WAR

Topic CPI In Pipeline



Happy Learning

THANK - YOU